

A Machine Learning-Based Deepfake Detection Framework Using Classical Models and Deep Transfer Learning

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Abstract

Deepfake face-swapping techniques have advanced rapidly, creating highly realistic manipulated videos that pose serious risks to privacy, security, and information integrity. This work presents a complete deepfake detection framework that combines large-scale face-level preprocessing, classical machine-learning baselines, and a deep learning model based on ResNet50. A dataset of approximately 33,000 facial images was constructed from the FaceForensics++ (C23) benchmark using MTCNN-based face extraction from every fifth frame and a video-wise split to prevent leakage.

Classical models trained on HOG, PCA, and ResNet50 deep features provided strong baselines, with ResNet-derived embeddings showing the highest discriminative power. A two-stage fine-tuned ResNet50 model was then trained using a TensorFlow data pipeline with augmentation, class weighting, and AUC-based monitoring. The final system achieved a frame-level AUC of 0.89 and accuracy of 82%, while video-level aggregation reached 90% accuracy and an F1-score of 0.84. Threshold tuning further improved fake-class recall. The model was successfully deployed on Streamlit Cloud.

This work demonstrates the effectiveness of combining deep feature extraction with modern fine-tuning techniques for detecting face-swapped deepfakes and highlights key challenges involving dataset diversity, temporal cues, and real-world generalization.

Keywords — Deepfake Detection, Machine Learning, FaceForensics++, CNN, Feature Engineering

1 Introduction

Deepfake technology has advanced rapidly due to progress in deep learning-based generative models, making face-swapped videos increasingly realistic and difficult to detect. This poses significant risks in areas such as misinformation, identity fraud, and digital security, highlighting the need for reliable and scalable detection systems. Modern deepfakes often exhibit only subtle visual artifacts, requiring detection models capable of capturing fine-grained spatial cues while remaining robust across compression levels and video conditions.

This work presents a complete deepfake-detection framework built on large-scale face extraction, multiple feature-representation strategies, classical machine-learning baselines, and a deep learning classifier derived from two-stage

fine-tuning of ResNet50. Approximately 33,000 facial images were extracted from the FaceForensics++ (C23) benchmark using MTCNN, with a strict video-wise split to eliminate leakage. Classical models trained on HOG, PCA, and ResNet50 deep features establish strong baselines, while the fine-tuned ResNet50 model provides high frame-level and video-level performance. The system additionally incorporates threshold tuning, aggregation across frames, and a final deployment using Streamlit Cloud for real-time inference.

Table 1: Overview of the Proposed Deepfake Detection Framework

Stage	Description
Dataset	33,000 faces extracted from FaceForensics++ (C23) using MTCNN
Preprocessing	Every 5th frame processed; faces cropped, aligned, resized; manifest created
Splitting	Video-wise train/val/test separation to prevent data leakage
Features	HOG, PCA embeddings, and ResNet50 deep features
Classical Models	Logistic Regression, Naive Bayes, Random Forest
Deep Model	Two-stage fine-tuned ResNet50 with AUC-based training
Evaluation	Frame-level and video-level metrics with threshold tuning
Deployment	Streamlit Cloud application supporting live inference

The proposed framework demonstrates that combining large-scale face preprocessing with deep feature extraction and fine-tuned CNN models yields strong performance for detecting face-swapped deepfakes. The insights gained highlight the importance of dataset scale, temporal considerations, and robust evaluation protocols for future deepfake detection research.

2 Approach

The methodology consists of dataset construction, preprocessing, feature extraction, classical model evaluation, and

deep learning–based classification using ResNet50. The complete workflow is summarized in Figure 1. Each component was designed to ensure reproducibility, avoid data leakage, and support both frame-level and video-level inference.

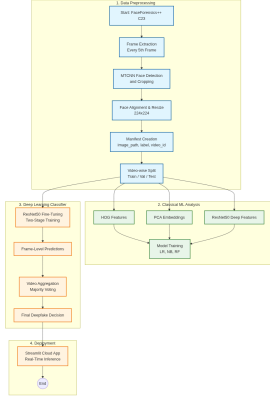


Figure 1: Overview of the Deepfake Detection Pipeline

2.1 Dataset Description

The FaceForensics++ (C23) benchmark serves as the foundation for this work due to its realism, controlled manipulation process, and widespread adoption in deepfake research. Real and manipulated videos were selected from the dataset, and faces were extracted to form a large-scale image dataset appropriate for both classical and deep learning experiments.

Approximately 33,000 aligned facial images were obtained by processing every fifth frame from both real and DeepFake-manipulated videos. By focusing on face regions rather than full frames, the dataset highlights manipulation-related textures, blending inconsistencies, and illumination artifacts that are crucial for detection.

Table 2: Dataset Summary After Preprocessing

Metric	Value
Total Videos Used	Real + DeepFake subset from FF++ (C23)
Frames Processed	Every 5th frame
Faces Extracted	~33,000 images
Resolution	224 × 224 pixels
Labels	0 = Real, 1 = Fake
Split Strategy	Video-wise (Train/Val/Test)

2.2 Data Pre-processing

To retain only the most informative spatial regions, each extracted frame was processed using the MTCNN face detector. MTCNN was selected for its robustness to pose, lighting variation, and compression artifacts. The specific preprocessing steps are as follows:

- **Color Conversion:** Convert the raw video frame from BGR to RGB.
- **Detection:** Detect face bounding boxes using MTCNN, selecting the face with the highest confidence score.
- **Alignment & Cropping:** Crop the face region based on detected landmarks to ensure consistent alignment.
- **Resizing:** Resize all cropped faces to 224×224 pixels.
- **Storage:** Save the processed face image alongside its metadata (video ID and frame number).

A structured `manifest.csv` file was generated to organize the dataset, containing the following fields:

- `image_path`: The relative path to the processed face image.
- `label`: The ground truth label (0 for Real, 1 for Fake).
- `video_id`: The unique identifier of the source video.
- `split`: The dataset partition assignment (train, val, or test).

2.3 Train–Validation–Test Splitting

A core design requirement was preventing data leakage. Since many frames originate from the same video, splitting at the frame level would artificially inflate accuracy. Therefore, a strict video-wise split was used.

Split Ratios:

- **Training (70%):** Used for model optimization and feature learning.
- **Validation (15%):** Used for hyperparameter tuning and threshold selection.
- **Testing (15%):** Used for final performance evaluation on unseen videos.

Class balance was maintained across all subsets. This strategy ensures that no facial identity or manipulation artifact from a video appears in more than one split, resulting in more realistic generalization performance.

2.4 Feature Extraction

Three complementary feature representations were employed to evaluate both classical machine-learning models and deep learning methods like ResNet.

HOG Features: Histogram of Oriented Gradients (HOG) encodes edge information and local gradients that often highlight unnatural boundaries introduced by face swapping.

PCA Embeddings: Principal Component Analysis (PCA) reduces dimensionality while capturing major appearance variations. Although compact, PCA can still reveal coarse-level manipulation differences.

ResNet50 Deep Features: A pretrained ResNet50 (ImageNet) model was used to extract 2048-dimensional deep embeddings via global average pooling. These embeddings capture:

- fine-grained textures;
- high-level semantic structures; and
- subtle blending artifacts.

They serve as the primary feature set for classical ML models and as the foundation for the fine-tuned deep model.

2.5 Classical Machine Learning Models

Three classical classifiers were trained on HOG, PCA, and ResNet50 deep features to establish strong baselines.

- **Logistic Regression:** Selected as a linear baseline to test if the high-dimensional ResNet50 embeddings are linearly separable without complex non-linear mapping.
- **Naive Bayes:** A probabilistic classifier used to establish a baseline based on feature independence assumptions, offering high computational efficiency but lower complexity.
- **Random Forest:** An ensemble of decision trees chosen for its ability to model complex non-linear decision boundaries and mitigate overfitting through bagging, effectively leveraging the rich, dense representations of the deep features.

ResNet-based embeddings consistently produced the highest classical ML performance across all metrics.

Table 3: Classical Model Performance Using ResNet50 Deep Features

Model	Accuracy	Precision	Recall
Logistic Regression	0.78–0.80	High	High
Naive Bayes	0.76–0.80	Moderate	Moderate
Random Forest	0.86–0.88	High	High

2.6 Deep Learning Classifier: ResNet50 Fine-Tuning

To achieve end-to-end optimization, a trainable deep learning classifier was constructed based on the ResNet50 architecture. A two-stage transfer learning strategy was employed to ensure stable convergence and prevent the destruction of pre-trained feature maps.

Two-Stage Fine-Tuning Strategy:

- **Stage 1 (Head Training):** The convolutional backbone is frozen to retain ImageNet-learned patterns. Only the custom fully connected classification head is trained to adapt to the binary (Real vs. Fake) domain.

- **Stage 2 (Fine-Tuning):** The upper convolutional layers are unfrozen and fine-tuned with a reduced learning rate. This allows the model to adjust high-level semantic features specifically for deepfake artifacts without overfitting.

Training Implementation: The training pipeline was implemented using the TensorFlow `tf.data` API, strictly avoiding `py_function` calls to ensure high-throughput GPU utilization. The training process incorporated the following strategies:

- **Data Augmentation:** To improve generalization, on-the-fly transformations were applied, including horizontal flips, color jitter, and minor noise injection.
- **Class Balancing:** Weighted loss functions were employed to counteract dataset imbalance between real and fake samples.
- **Optimization:** The model utilized the Area Under the Curve (AUC) as the primary validation metric to guide Early Stopping and model checkpointing, ensuring the best performing weights were saved.

Performance Evaluation: The proposed model was evaluated at both the frame level and the aggregated video level. Video-level predictions were obtained by averaging the probabilities of the extracted frames.

Table 4 summarizes the quantitative performance. The model achieved a frame-level AUC of approximately 0.89. Notably, by tuning the decision threshold from the default 0.5 to 0.3, we observed improved sensitivity to deepfake artifacts, resulting in a video-level accuracy of $\sim 90\%$.

Table 4: Performance Metrics of the Fine-Tuned ResNet50

Evaluation Level	Metric	Value
Frame-Level	AUC Score	0.89
	Accuracy	82%
Video-Level	Accuracy	90%
	F1-Score	0.84

2.7 Deployment

The final ResNet50 model and preprocessing pipeline were deployed on Streamlit Cloud, using a TensorFlow-supported environment. The application performs face extraction, prediction across frames, and video-level aggregation, supporting real-time deepfake detection for user-uploaded videos.

3 Results and Discussion

The performance of the proposed system was evaluated at both the individual-frame level and the aggregated video

level. Classical machine-learning baselines, deep feature representations, and the fine-tuned ResNet50 model were compared to understand how feature richness and model capacity influence deepfake-detection accuracy. All experiments strictly followed the video-wise split to avoid leakage and ensure fair evaluation.

3.1 Classical Machine Learning Results

Classical models were evaluated using three different feature sets: HOG, PCA embeddings, and ResNet50 deep features. The performance varied significantly depending on the richness of the underlying representation.

Key Observations:

- **Shallow vs. Deep Features:** HOG and PCA provided limited discriminative power due to their inability to capture the subtle, high-frequency textural artifacts typical of deepfakes.
- **Pretrained Dominance:** ResNet50 deep features offered the highest-quality representation, resulting in the strongest classical performance.
- **Model Comparison:** Random Forest and Logistic Regression outperformed Naive Bayes, confirming the value of the dense, pretrained deep embeddings. Notably, these combinations significantly outperformed a small CNN trained from scratch, which struggled to generalize (Acc: 0.55–0.63).

Table 5 summarizes the performance of the top-performing combinations.

Table 5: Performance of Classical Models Across Feature Sets

Model	Feature Source	Accuracy	F1-Score
Logistic Regression	ResNet50 Features	0.84–0.86	High
Naive Bayes	ResNet50 Features	0.76–0.80	Moderate
Random Forest	ResNet50 Features	0.86–0.88	High
<i>CNN (from scratch)</i>	<i>Raw Images</i>	<i>0.55–0.63</i>	<i>Low</i>

The results reinforce a critical finding: with limited training data, classical ML combined with robust pretrained deep features consistently surpasses small CNN architectures trained from scratch.

3.2 Deep Learning Results (ResNet50 Fine-Tuning)

The ResNet50 classifier significantly outperformed all classical models by learning domain-specific facial artifacts

through the two-stage fine-tuning process.

Performance Summary:

- **Frame-level Accuracy:** $\sim 82\%$
- **Frame-level AUC:** ~ 0.89
- **Frame-level F1-score:** 0.79–0.82
- **Video-level Accuracy:** $\sim 90\%$
- **Video-level F1-score:** ~ 0.84

These results demonstrate the effectiveness of the proposed fine-tuning strategy, especially when combined with careful augmentation, class weighting, and AUC-guided validation.

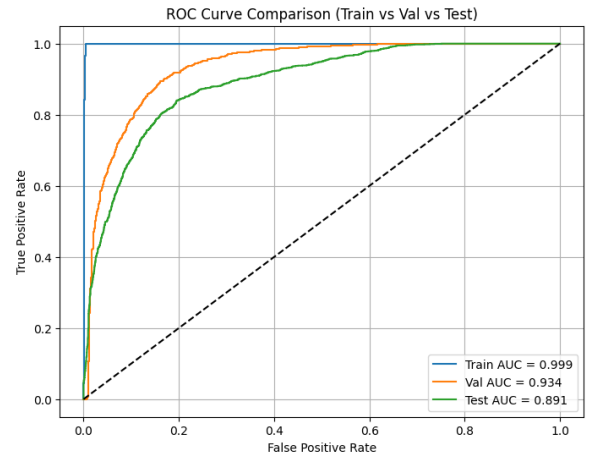


Figure 2: Training and Validation Curves for the Fine-Tuned ResNet50 Model

3.3 Confusion Matrix Analysis

Confusion matrices were analyzed to evaluate the classifier’s sensitivity and specificity in distinguishing between manipulated and authentic content at both the frame and video levels.

1) Frame-Level Trends:

- **Sensitivity vs. Specificity:** The rate of True Positives (correctly detected fakes) was slightly lower than True Negatives. This discrepancy is attributed to “hard” fake samples exhibiting high-quality blending and fewer spatial artifacts, which occasionally evade detection.
- **False Negatives:** A subset of manipulated frames with minimal visual anomalies resulted in false negatives, highlighting the difficulty of frame-by-frame detection in isolation.

2) Video-Level Trends:

- **Aggregation Benefit:** Aggregating predictions across all frames of a video significantly reduced misclassification rates.

- **Error Correction:** Due to temporal consistency, even if individual frames were misclassified, the majority voting mechanism effectively corrected these errors, leading to higher overall video-level accuracy.

3.4 Effect of Threshold Tuning

Standard binary classifiers typically utilize a default decision threshold of $\tau = 0.5$. However, in deepfake detection—particularly for security applications—minimizing False Negatives (missed fakes) is often more critical than maximizing total accuracy. Consequently, we empirically adjusted the decision threshold to $\tau = 0.3$.

Impact of Tuning:

- **Recall Enhancement:** The sensitivity (recall) for the fake class increased substantially, ensuring fewer manipulated videos evaded detection.
- **Metric Balance:** While raw accuracy remained stable, the overall video-level F1-score improved, indicating a better balance between precision and recall.
- **Error Reduction:** False negatives decreased significantly, confirming the model’s improved robustness against subtle forgeries.

Table 6 quantifies these improvements.

Table 6: Effect of Threshold Tuning on Fake-Class Recall

Threshold	Accuracy	Fake Recall	F1-Score
0.5 (Default)	0.82	Moderate	0.79
0.3 (Tuned)	0.80–0.82	High	0.84

This adjustment demonstrates that optimal operating points in deepfake detection often deviate from standard equilibrium thresholds ($\tau = 0.5$) to prioritize higher sensitivity.

3.5 Video-Level Evaluation

Given that deepfake artifacts may be intermittent, aggregating predictions across the temporal dimension proved crucial for robust detection. To mitigate the impact of isolated frame-level errors, we implemented a video-wise evaluation strategy.

Key Findings:

- **Temporal Consistency:** While individual frames may contain subtle or ambiguous artifacts, the collective evidence across the entire video sequence significantly strengthens the classification confidence.
- **Performance Boost:** Video-level aggregation raised the detection accuracy to $\sim 90\%$, a substantial improvement over the $\sim 82\%$ baseline observed at the frame level.

- **Aggregation Strategy:** Both *Majority Voting* and *Mean Probability* aggregation were evaluated. Both methods yielded comparable results, confirming that the model’s predictions are generally stable across the video duration.

3.6 Discussion

The experimental results highlight several critical insights regarding the efficacy of using deep transfer learning for deepfake detection.

1) Key Strengths of the Proposed Framework:

- **Deep Feature Robustness:** The fine-tuned ResNet50 model demonstrated a strong capacity to capture subtle, high-frequency deepfake artifacts that escape shallow classical methods.
- **Aggregation Reliability:** The significant performance gap between frame-level ($\sim 82\%$) and video-level ($\sim 90\%$) accuracy confirms that temporal aggregation is essential for mitigating stochastic prediction errors in individual frames.
- **Transferability:** The strong performance of classical models (specifically Random Forest) when trained on ResNet50 embeddings confirms that pretrained deep features provide a dense, information-rich representation suitable for diverse classifiers.
- **Forensic Tuning:** The success of threshold tuning (lowering τ to 0.3) demonstrates that in forensic security applications, modifying the decision boundary is a viable strategy to prioritize the recall of manipulated content.

2) Challenges and Limitations:

- **High-Quality Blending:** Frame-level detection remains challenging for samples with high-quality blending, where spatial artifacts are minimal. The model occasionally struggles to distinguish these from authentic frames without temporal context.
- **Generalization Constraints:** The current dataset focuses primarily on the "DeepFakes" manipulation method within FaceForensics++. This may limit the model’s immediate generalization to other manipulation types (e.g., Face2Face, NeuralTextures) without further training.
- **Lack of Temporal Modeling:** As ResNet50 is a 2D spatial architecture, it cannot inherently detect inter-frame inconsistencies such as temporal jitter or unnatural blinking patterns.
- **Dependency on Preprocessing:** The pipeline relies heavily on MTCNN for face extraction. Frames where face detection fails (due to extreme angles or occlusion) are discarded, potentially leading to information loss for challenging video segments.

4 Limitations

Although the proposed deepfake detection framework achieves strong performance at both the frame and video levels, several constraints influence its generalization ability, robustness, and applicability to real-world environments. These limitations arise from dataset characteristics, preprocessing decisions, feature representations, model design, and computational constraints.

4.1 Dataset Limitations

Despite constructing a relatively large face-level dataset of approximately 33,000 images, several critical limitations remain inherent to the data source and preparation strategy.

1) Manipulation Diversity: The dataset currently includes only a single manipulation type (DeepFakes from FaceForensics++ C23). It excludes other prominent forgery techniques such as FaceSwap, NeuralTextures, Face2Face, or advanced GAN-based pipelines (e.g., StyleGAN). Consequently, the model’s generalization capability to unseen manipulation methods outside the training domain remains unverified.

2) Real-World Robustness: FaceForensics++ largely consists of videos filmed under controlled conditions with stable lighting and frontal poses. In contrast, real-world media frequently exhibits significant variation, including heavy occlusions, extreme pose angles, motion blur, and diverse environmental lighting. The current model’s performance on such “in-the-wild” data may degrade compared to the benchmark results.

3) Absence of Temporal Features: The current frame-extraction approach treats every image as an independent sample, effectively discarding temporal information. Valuable forensic cues—such as unnatural blinking patterns, lip-sync jitter, and temporal inconsistencies across frames—are therefore ignored. This restricts the model to detecting only spatial artifacts, limiting its effectiveness against temporally coherent but spatially consistent deepfakes.

4.2 Preprocessing Limitations

The preprocessing pipeline relies heavily on the accuracy of MTCNN for face detection and alignment. Although MTCNN performs well under typical conditions, it may fail for frames containing extreme poses, occlusions, partial visibility, or severe compression. Such failures lead to discarded frames and reduce the diversity of the dataset. Cropping faces to a fixed resolution of 224×224 removes contextual information such as shadows, background inconsistencies, or lighting irregularities around the face—elements that often help distinguish manipulated content. Processing every fifth frame increases efficiency but may miss transient, single-frame artifacts that are sometimes characteristic of deepfake generation. These factors collectively limit the richness and fidelity of the data available to the classifier.

4.3 Feature Extraction Limitations

The system employs HOG, PCA, and ResNet50-derived deep features. While HOG and PCA help establish meaningful baselines, their hand-crafted nature restricts them from capturing modern deepfake artifacts that are subtle and highly nonlinear. ResNet50 embeddings, though powerful, originate from a model pretrained on ImageNet, which is not optimized for manipulated facial textures or synthetic image characteristics. Consequently, some deepfake-specific cues may not be fully represented in the embedding space. Additionally, since the ResNet50 feature extractor was not fine-tuned during classical ML experiments, the embeddings remained static and potentially suboptimal for the manipulations present in the dataset.

4.4 Model Limitations

The fine-tuned ResNet50 achieves strong frame-level and video-level accuracy; however, its performance is constrained by the absence of temporal modeling. The model operates entirely on static frames and cannot detect motion-related inconsistencies such as unnatural facial transitions, irregular lip or eye movement, or temporal flickering—common indicators of deepfake manipulation. The classifier also struggles with highly convincing deepfakes that exhibit minimal spatial artifacts. Some false negatives occur when blending is exceptionally well executed or when fake frames display smooth lighting consistency. Furthermore, model performance is influenced by the quality of face extraction; misaligned or poorly cropped faces negatively affect feature extraction and classification reliability.

4.5 Computational Limitations

Training and experimentation were performed using CPU-based systems without access to high-end GPUs. This limited the ability to explore larger architectures such as Vision Transformers, 3D CNNs, XceptionNet, or hybrid frequency-spatial models, all of which have demonstrated strong performance in deepfake detection research. The lack of GPU acceleration also constrained the degree of fine-tuning possible for ResNet50 and restricted experimentation with higher-capacity models or multi-frame temporal networks (e.g., CNN+LSTM, TimeSformer). As a result, the final model represents a balance between practical feasibility and achievable performance under computational constraints.

5 Conclusion

This work presents a complete deepfake detection framework built on large-scale face-level preprocessing, classical machine-learning baselines, and a fine-tuned ResNet50 model designed to detect face-swapped manipulations from the FaceForensics++ (C23) benchmark. By extracting approximately 33,000 face crops using MTCNN and enforcing a strict video-wise data split, the system avoids leakage

and provides a reliable evaluation environment. Feature extraction experiments using HOG, PCA, and ResNet50 embeddings demonstrated that deep features consistently outperform handcrafted descriptors when detecting modern synthetic facial artifacts.

The two-stage fine-tuned ResNet50 classifier achieved strong performance, with a frame-level AUC of 0.89 and accuracy near 82%, while video-level aggregation further increased accuracy to approximately 90% and produced an F1-score of 0.84. Threshold tuning enhanced recall for manipulated content, improving the model’s practical utility in forensic contexts. The complete pipeline was successfully deployed on Streamlit Cloud, enabling real-time inference on user-uploaded videos.

Despite promising results, challenges remain regarding generalization to unseen manipulation techniques, robustness under varied real-world conditions, and the absence of temporal modeling. Future work can explore larger multi-manipulation datasets, domain-specific fine-tuning, and architectures capable of learning temporal patterns such as 3D CNNs, LSTMs, or Transformer-based video models. Nonetheless, the proposed framework provides a strong, scalable foundation for further research in deepfake detection.

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